

— Evaluation and Visualization Report

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1. Project Description

Hostel Hunter is a fuzzy inference system designed to recommend hostels according to traveller preferences. The system evaluates hostels using multiple criteria such as price, distance from city centre, safety, cleanliness, and social vibe. Instead of using rigid binary logic, the system applies fuzzy logic principles to produce gradual and realistic recommendations.

2. Motivation

Traditional recommendation systems often rely on hard thresholds, which can produce unrealistic decisions. For example, a hostel that is 3.1 km away may be rejected while a hostel 2.9 km away is accepted. Fuzzy logic solves this issue by allowing partial memberships and smooth transitions between categories such as low price, medium price, or high safety. This makes the recommendation system more human-like and flexible.

3. Input Variables

The fuzzy system uses five main input variables:

- Price per night (€)
- Distance from city centre (km)
- Safety rating (1–10)
- Cleanliness rating (1–10)
- Social vibe rating (1–10)

4. Example Default User Preferences

Default example user preferences used in the evaluation:

```
HOSTEL HUNTER - Interactive Mode

Enter your travel preferences:

Max budget per night (€) [5-120, default=40]: 50
Max distance from centre (km) [0.5-15, default=3]: 3
Min safety rating (1-10) [1-10, default=6]: 6
Min cleanliness rating (1-10) [1-10, default=7]: 7
Social vibe importance (1-10) [1-10, default=5]: 5


HOSTEL HUNTER - FUZZY LOGIC RECOMMENDATION SYSTEM

UserPreferences(budget=€50.0, distance≤3.0km, safety≥6.0, cleanliness≥7.0, social=5.0/10)


RANK   HOSTEL                                CITY      PRICE    DIST    SCORE    TIER
#1     The Wanderer's Den                    Prague    € 22     0.8km   90.6%    ★ Best Match
       [██████████░░░░░░░░░░░░░░░░] fuzzy=0.85 pref=0.97
#2     Urban Nest                            Barcelona € 45     0.5km   90.2%    ★ Best Match
       [██████████░░░░░░░░░░░░░░░░] fuzzy=0.85 pref=0.96
#3     Central Slumber                       Vienna    € 38     0.3km   89.8%    ★ Best Match
       [██████████░░░░░░░░░░░░░░░░] fuzzy=0.86 pref=0.95
#4     Skyline Crash Pad                     Amsterdam € 55     1.0km   89.0%    ★ Best Match
       [██████████░░░░░░░░░░░░░░░░] fuzzy=0.86 pref=0.93
#5     Bunk & Brew                           Berlin    € 30     2.1km   88.6%    ★ Best Match
       [██████████░░░░░░░░░░░░░░░░] fuzzy=0.82 pref=0.97
```

Fuzzy Rules fired for "The Wanderer's Den":

```
R01 [ ]  $\mu=0.800 \rightarrow \text{output}=0.90$   
IF price LOW AND safety HIGH  $\rightarrow$  recommendation HIGH  
R02 [ ]  $\mu=0.100 \rightarrow \text{output}=0.88$   
IF price MEDIUM AND safety HIGH AND cleanliness HIGH  $\rightarrow$  HIGH  
R03 [ ]  $\mu=1.000 \rightarrow \text{output}=0.85$   
IF distance NEAR AND safety HIGH  $\rightarrow$  recommendation HIGH  
R04 [ ]  $\mu=0.667 \rightarrow \text{output}=0.87$   
IF cleanliness HIGH AND safety HIGH  $\rightarrow$  HIGH  
R05 [ ]  $\mu=0.800 \rightarrow \text{output}=0.80$   
IF social HIGH AND price LOW  $\rightarrow$  HIGH (backpacker vibe)  
R06 [ ]  $\mu=0.100 \rightarrow \text{output}=0.82$   
IF cleanliness HIGH AND distance NEAR AND price MEDIUM  $\rightarrow$  HIGH
```

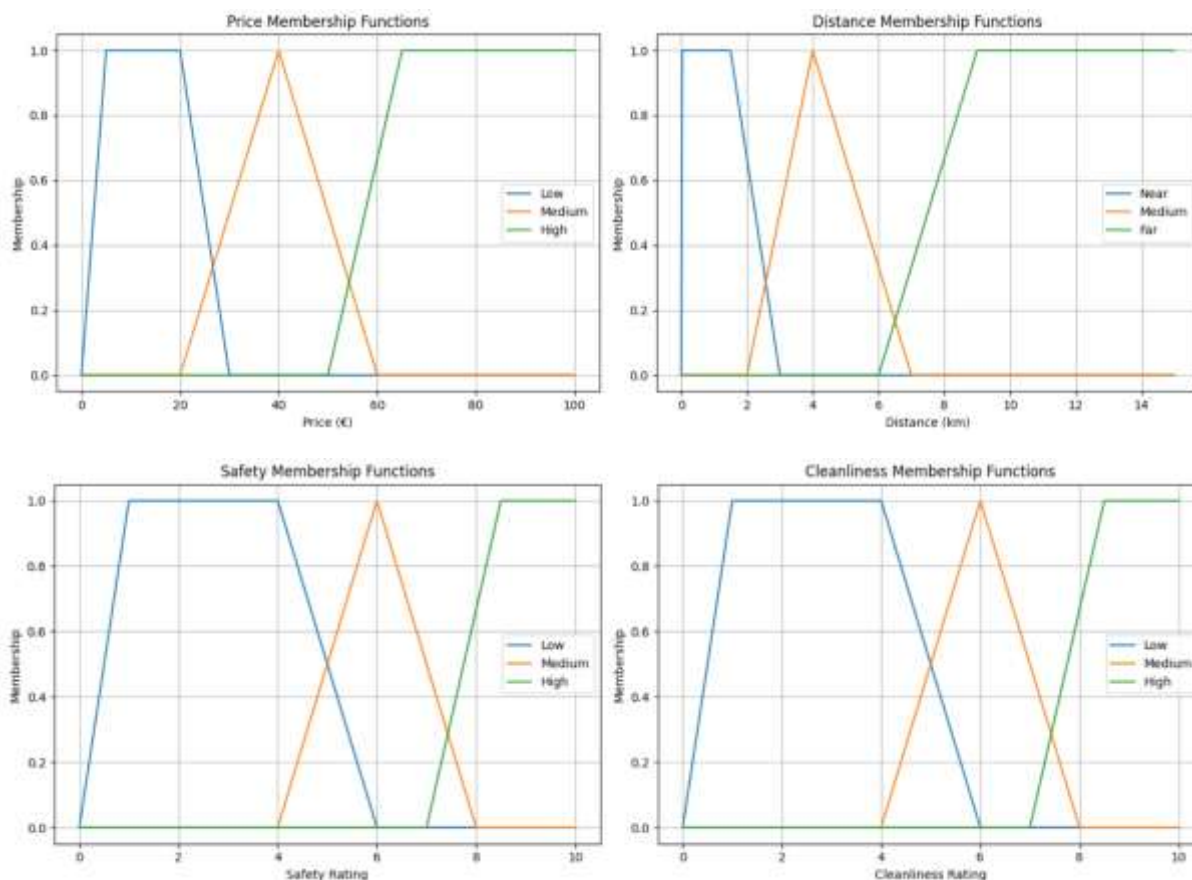
Membership values for "The Wanderer's Den" (Price=€22, Dist=0.8km, Safety=8.5, Clean=8.0, Social=9.0):

Price	low=0.800	medium=0.100	high=0.000
Distance	near=1.000	medium=0.000	far=0.000
Safety	low=0.000	med=0.000	high=1.000
Cleanliness	low=0.000	med=0.000	high=0.667
Social	low=0.000	med=0.000	high=1.000

Defuzzification: Centre-of-Gravity | Blend: 55% fuzzy + 45% preference match

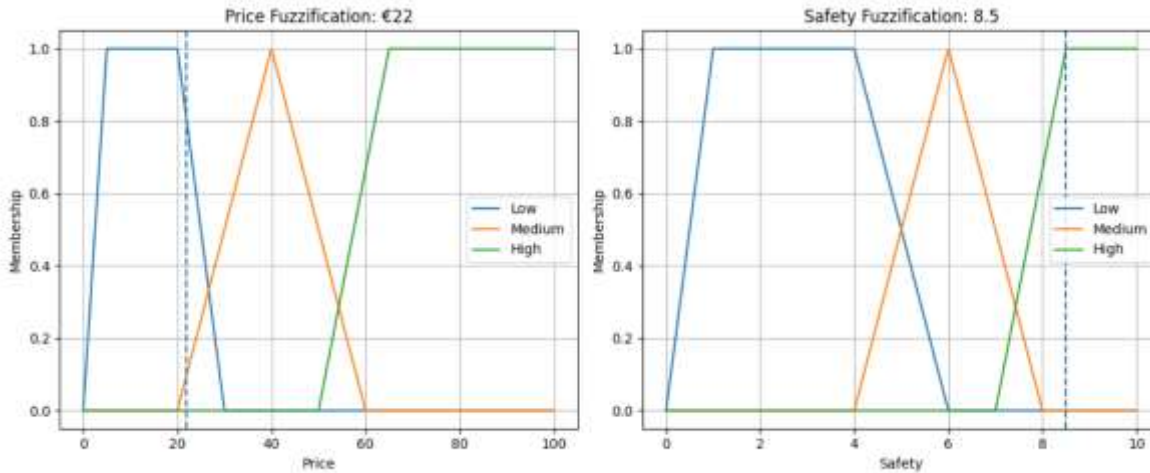
5. Membership Function Graphs

The membership function graphs demonstrate overlapping fuzzy sets for price, distance, safety, and cleanliness. These overlapping regions prove that hostel attributes can partially belong to multiple categories simultaneously, which is a core principle of fuzzy logic.



6. Fuzzification Visualization

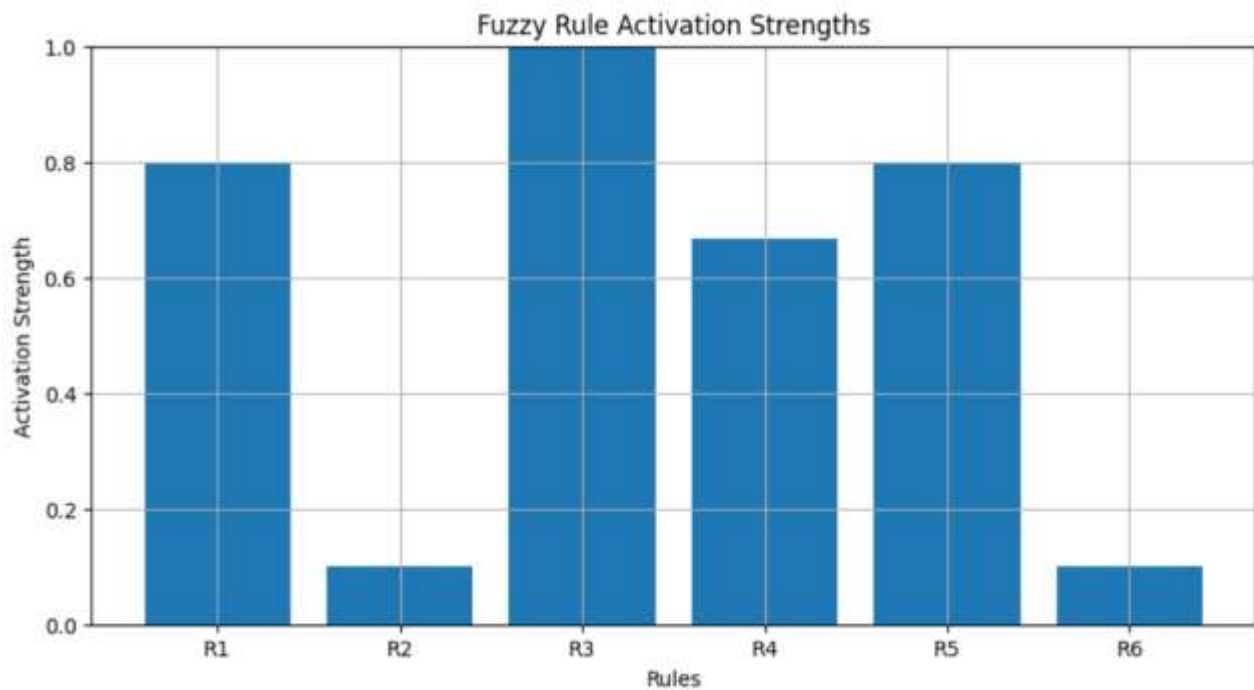
The fuzzification graphs demonstrate how crisp hostel inputs are converted into fuzzy membership values. For example, a hostel price of €22 belongs strongly to the LOW price category while also partially belonging to the MEDIUM category.



7. Rule Activation Analysis

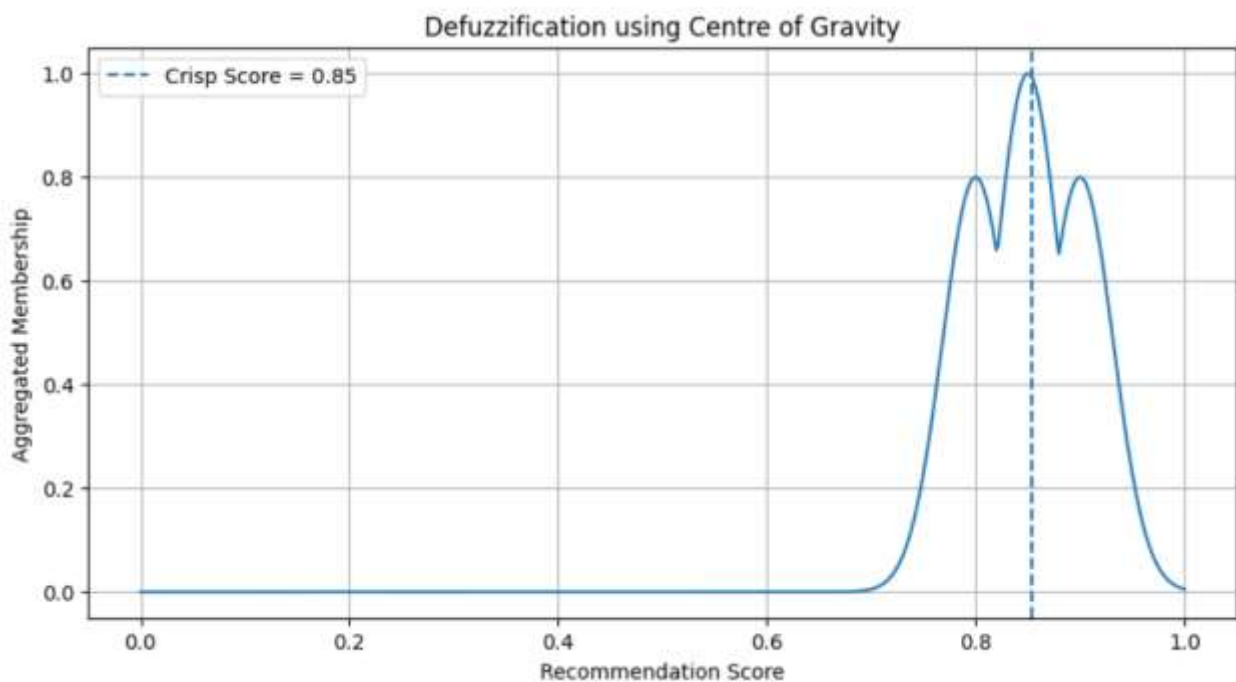
The rule activation graph proves that multiple fuzzy rules are triggered simultaneously with different activation strengths. Stronger rules contribute more heavily to the final recommendation score.

1. Rule 1: Cheap + safe = great
($\min(m[\text{"price_low"}], m[\text{"safety_high"}])$, 0.90, "IF price LOW AND safety HIGH → recommendation HIGH"),
2. Rule 2: Cheap + medium safety = decent
($\min(m[\text{"price_low"}], m[\text{"safety_med"}])$, 0.70, "IF price LOW AND safety MEDIUM → recommendation MEDIUM-HIGH"),
3. Rule 3: Expensive + far = avoid
($\min(m[\text{"price_high"}], m[\text{"dist_far"}])$, 0.10, "IF price HIGH AND distance FAR → recommendation LOW"),
4. Rule 4: Medium price + high safety + clean = excellent
($\min(m[\text{"price_medium"}], m[\text{"safety_high"}], m[\text{"clean_high"}])$, 0.88, "IF price MEDIUM AND safety HIGH AND cleanliness HIGH → HIGH"),
5. Rule 5: Close + safe = good
($\min(m[\text{"dist_near"}], m[\text{"safety_high"}])$, 0.85, "IF distance NEAR AND safety HIGH → recommendation HIGH"),
6. Rule 6: Far + expensive = bad
($\min(m[\text{"dist_far"}], m[\text{"price_high"}])$, 0.15, "IF distance FAR AND price HIGH → recommendation LOW"),
7. Rule 7: Clean + safe = reliable
($\min(m[\text{"clean_high"}], m[\text{"safety_high"}])$, 0.87, "IF cleanliness HIGH AND safety HIGH → HIGH"),
8. Rule 8: Dirty = low priority
($m[\text{"clean_low"}]$, 0.20, "IF cleanliness LOW → recommendation LOW"),
9. Rule 9: Unsafe = disqualify
($m[\text{"safety_low"}]$, 0.05, "IF safety LOW → recommendation VERY LOW"),
10. Rule 10: Budget + medium clean + near = solid
($\min(m[\text{"price_low"}], m[\text{"clean_med"}], m[\text{"dist_near"}])$, 0.72, "IF price LOW AND cleanliness MEDIUM AND distance NEAR → MEDIUM-HIGH"),
11. Rule 11: Social + cheap = backpacker dream
($\min(m[\text{"social_high"}], m[\text{"price_low"}])$, 0.80, "IF social HIGH AND price LOW → HIGH (backpacker vibe)"),



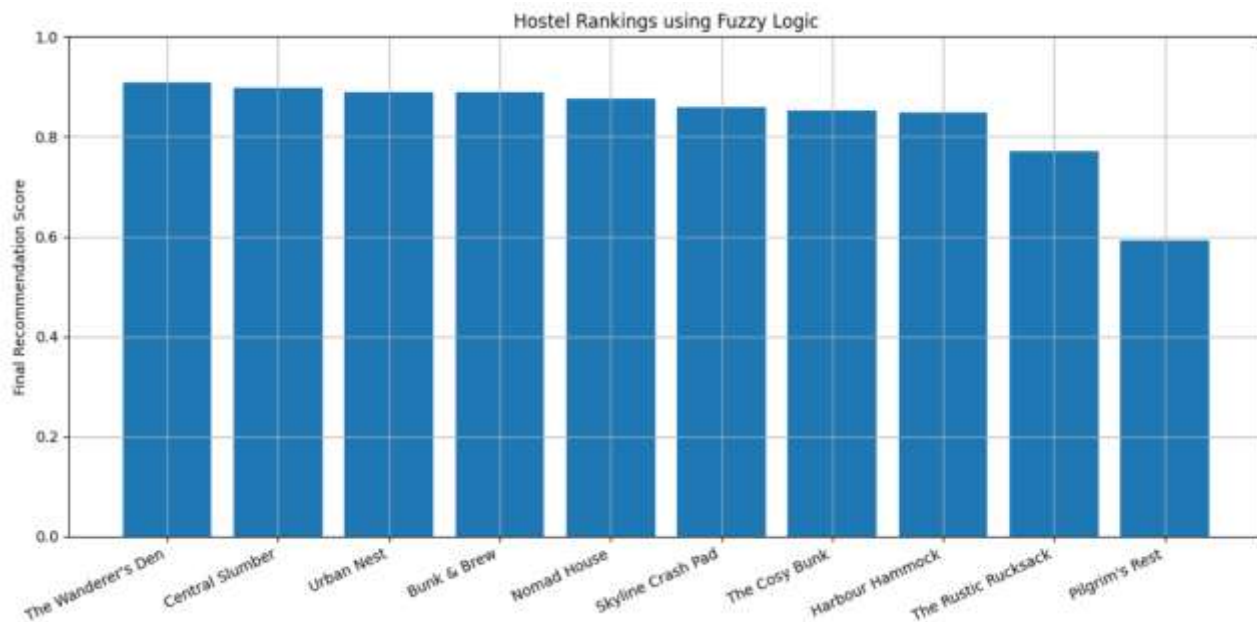
8. Defuzzification Analysis

The defuzzification graph demonstrates how multiple fuzzy outputs are aggregated into a single crisp recommendation score using the Centre-of-Gravity method.



9. Hostel Ranking Visualization

The hostel ranking graph demonstrates that the fuzzy system produces meaningful recommendation differences between hostels. Higher-quality hostels receive stronger recommendation scores.

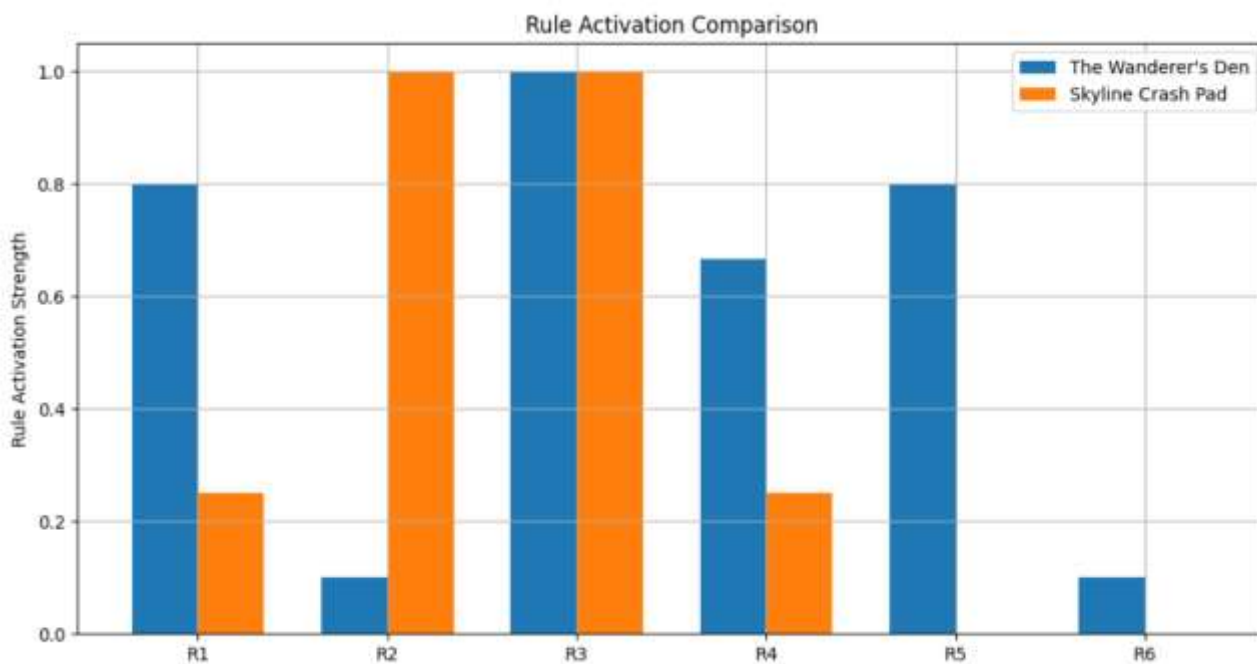


10. Comparison of Two Different Hotels

This comparison demonstrates that different hostels activate different fuzzy rules depending on their characteristics.

```
hotel1 = HOSTELS[0]
hotel2 = HOSTELS[2]

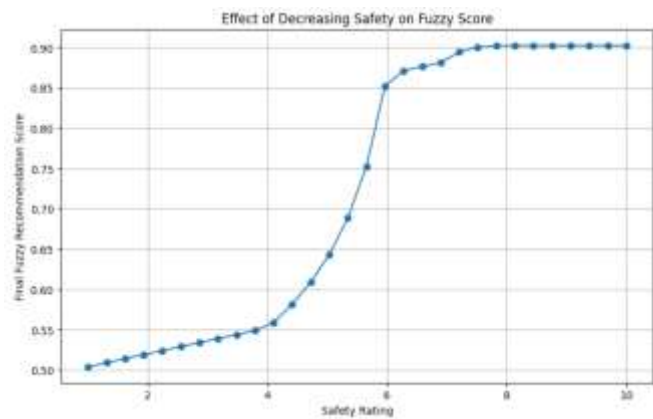
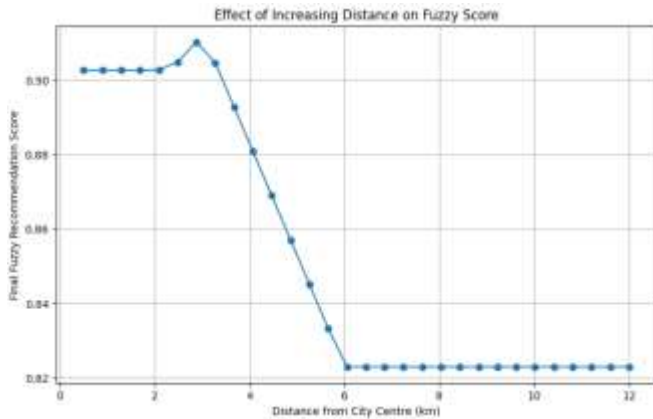
prefs = UserPreferences(
    budget=40,
    max_distance=3,
    min_safety=7,
    min_cleanliness=7,
    social_importance=6
)
```



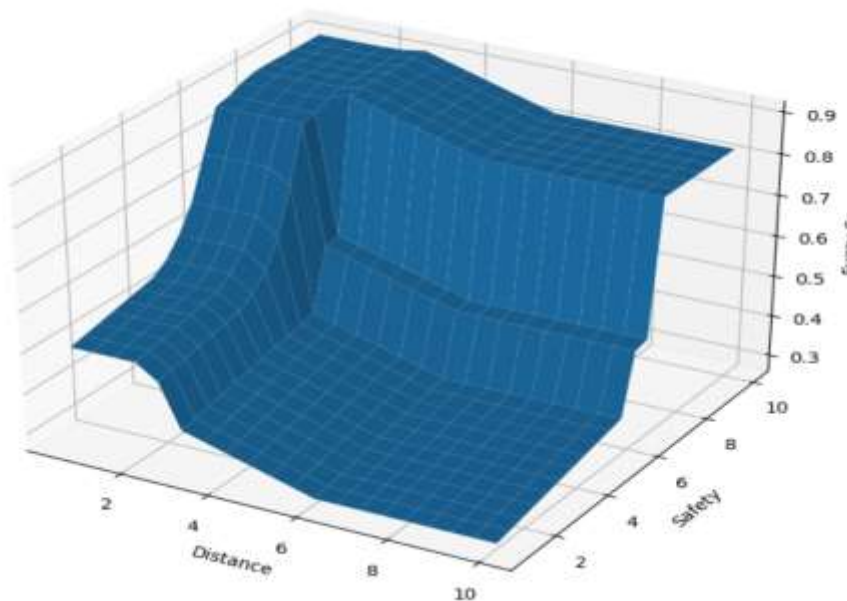
11. Input variables vs Fuzzy Score

This visualization proves monotonic fuzzy behaviour. As hostel distance from the city centre increases, the fuzzy recommendation score gradually decreases. The smooth transition confirms correct fuzzy inference behaviour.

Right side graph demonstrates that increasing safety ratings improve the final fuzzy recommendation score. The smooth nonlinear transitions validate gradual fuzzy reasoning.



3D Surface of Fuzzy Recommendation System



15. Summary and Conclusion

The evaluation results confirm that the Hostel Hunting system correctly implements fuzzy logic principles for hostel recommendation. The visualizations demonstrate membership overlap, fuzzification, simultaneous rule activation, aggregation, and defuzzification. Sensitivity analyses additionally prove that recommendation scores respond logically and smoothly to changing hostel attributes such as distance and safety.

The project successfully demonstrates the advantages of fuzzy logic in recommendation systems by providing gradual, interpretable, and realistic hostel recommendations instead of rigid binary decisions.